HTTP in Detail Report

1. Objective

The objective of this report was to understand the HTTP protocol in depth, including requests, responses, methods, headers, status codes, cookies, and sessions.

Scope of understanding:

- HTTP requests and responses
- HTTP methods and headers
- Status codes, cookies, and sessions

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2. Concept and Understanding

HTTP (HyperText Transfer Protocol) is the foundation of communication on the web. It defines how clients and servers exchange requests and responses.

HTTP and HTTPS

Protocol	Security
HTTP	Unencrypted
HTTPS	Encrypted using TLS

HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Submit data
PUT	Update data
DELETE	Remove data

Headers are metadata sent with requests and responses. They provide additional information about the communication between the client and the server.

Common Status Codes

Status Code	Meaning
200	OK
301	Moved Permanently
403	Forbidden
404	Not Found
500	Internal Server Error

Cookies and Sessions

Concept	Purpose
Cookies	Small pieces of data stored on the client
Sessions	State maintained on the server

Parameters are data sent in the URL for GET requests or in the request body for POST requests.

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3. Practical Work

The TryHackMe *HTTP in Detail* room was completed. The practical work included the following activities:

- Learned the structure of HTTP requests and responses
- Identified different HTTP methods
- Analyzed request and response headers
- Understood status-code meanings
- Explored cookies and session management
- Used Developer Tools to inspect real HTTP traffic

4. Tools Used

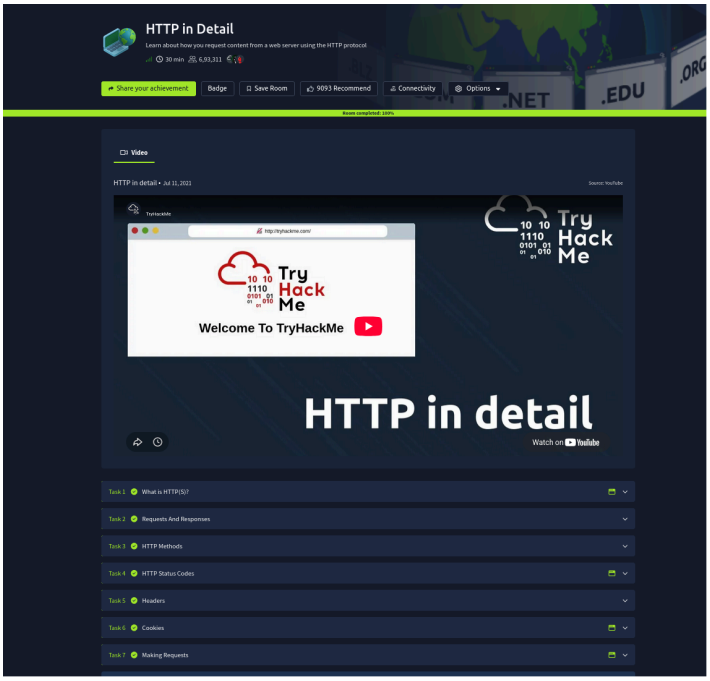
- TryHackMe
- Google Chrome
- Developer Tools (F12)

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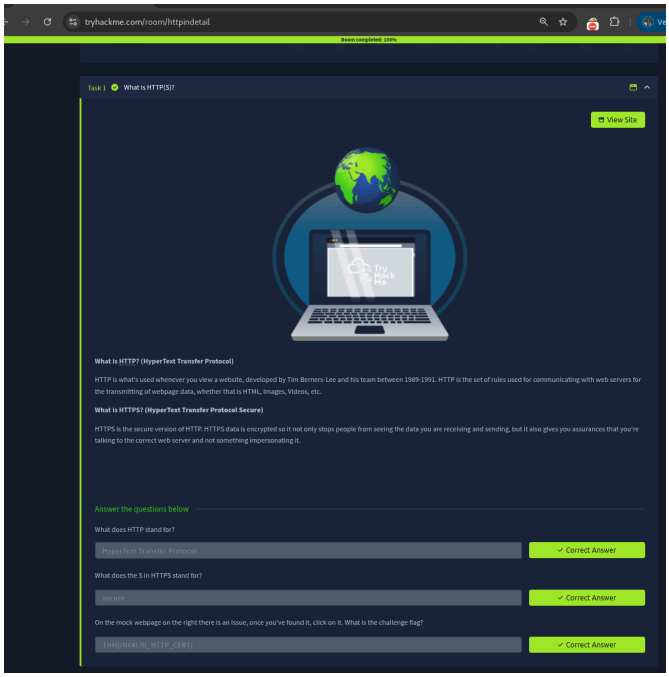
5. Evidence

The following numbered placeholders identify where the supporting screenshots can be inserted later.

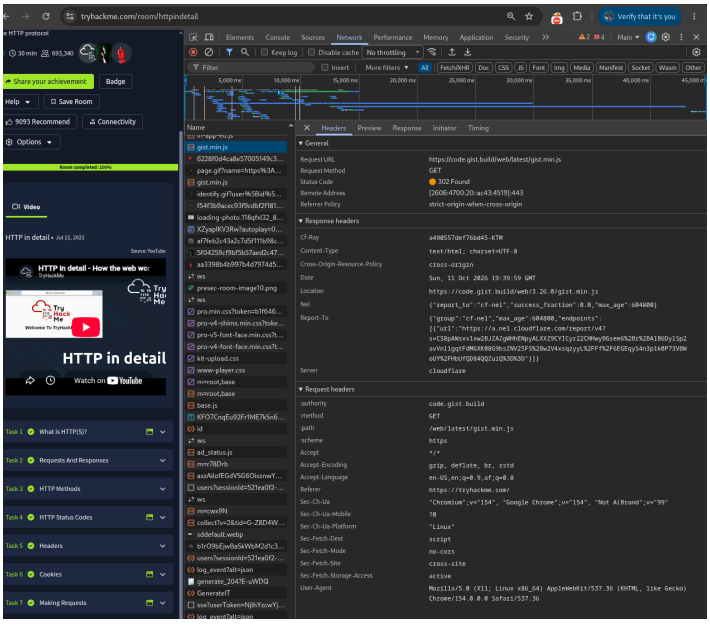
1. **HTTP in Detail room completion**



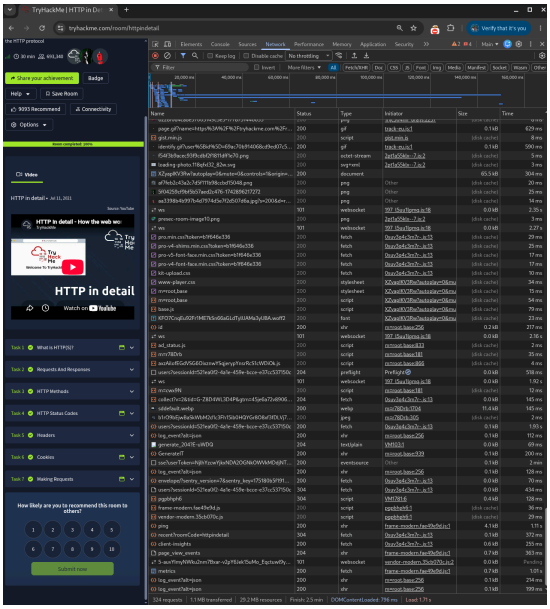
2. **Correct answers in the room**



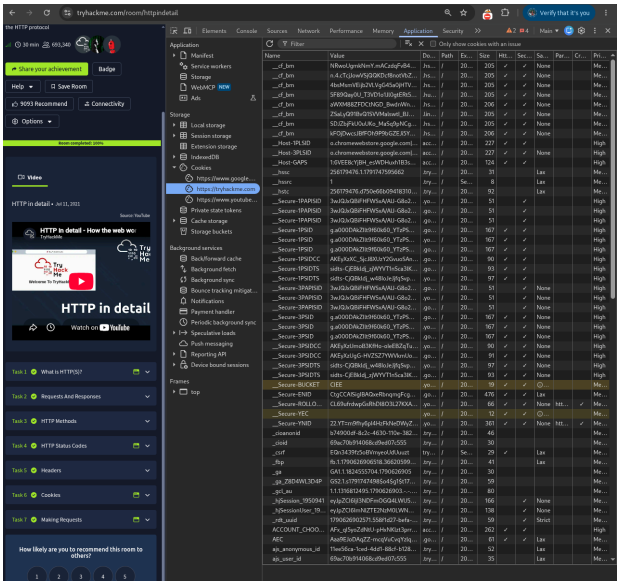
3. **GET request headers**



4. **Status codes**



5.Cookies and session data



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6. Result

The work successfully demonstrated an understanding of the following aspects of HTTP:

- The full structure of HTTP requests and responses
- Differences between HTTP methods
- The role of headers in communication
- How cookies and sessions work
- How to inspect these elements in real time

7. Key Learnings

- HTTP is a stateless protocol.
- Cookies and sessions help maintain state.
- Headers carry important metadata.
- Status codes communicate the outcome of a request.
- Developer Tools reveal HTTP activity in real time.

Conclusion

Understanding HTTP is practically important because it explains how web communication works and provides a foundation for analyzing requests, responses, and security-related behavior in real-world web environments.
